Class08: Breast Cancer Mini Project

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Background

In today's class we will be employing all the R techniques for data analysis that we have learned thus far - including the machine learning methods of clustering and PCA - to analyze real breast cancer biopsy data.

The data is in CSV format:

```
wisc.df <- read.csv("WisconsinCancer.csv", row.names=1)
```

We can have a peak at the data

```
head(wisc.df, 4)
```

```
         diagnosis radius_mean texture_mean perimeter_mean area_mean
842302           M       17.99        10.38         122.80    1001.0
842517           M       20.57        17.77         132.90    1326.0
84300903         M       19.69        21.25         130.00    1203.0
84348301         M       11.42        20.38          77.58     386.1
         smoothness_mean compactness_mean concavity_mean concave.points_mean
```

```
842302          0.11840          0.27760         0.3001             0.14710
842517          0.08474          0.07864         0.0869             0.07017
84300903        0.10960          0.15990         0.1974             0.12790
84348301        0.14250          0.28390         0.2414             0.10520
         symmetry_mean fractal_dimension_mean radius_se texture_se perimeter_se
842302          0.2419                0.07871    1.0950     0.9053        8.589
842517          0.1812                0.05667    0.5435     0.7339        3.398
84300903        0.2069                0.05999    0.7456     0.7869        4.585
84348301        0.2597                0.09744    0.4956     1.1560        3.445
         area_se smoothness_se compactness_se concavity_se concave.points_se
842302    153.40      0.006399        0.04904      0.05373           0.01587
842517     74.08      0.005225        0.01308      0.01860           0.01340
84300903   94.03      0.006150        0.04006      0.03832           0.02058
84348301   27.23      0.009110        0.07458      0.05661           0.01867
         symmetry_se fractal_dimension_se radius_worst texture_worst
842302       0.03003             0.006193        25.38         17.33
842517       0.01389             0.003532        24.99         23.41
84300903     0.02250             0.004571        23.57         25.53
84348301     0.05963             0.009208        14.91         26.50
         perimeter_worst area_worst smoothness_worst compactness_worst
842302            184.60     2019.0           0.1622            0.6656
842517            158.80     1956.0           0.1238            0.1866
84300903          152.50     1709.0           0.1444            0.4245
84348301           98.87      567.7           0.2098            0.8663
         concavity_worst concave.points_worst symmetry_worst
842302            0.7119               0.2654         0.4601
842517            0.2416               0.1860         0.2750
84300903          0.4504               0.2430         0.3613
84348301          0.6869               0.2575         0.6638
         fractal_dimension_worst
842302                   0.11890
842517                   0.08902
84300903                 0.08758
84348301                 0.17300
```

Q1. How many observations are in this dataset?

```r
nrow(wisc.df)
```

```
[1] 569
```

Q2. How many of the observations have a malignant diagnosis

```r
table(wisc.df$diagnosis)
```

```
  B   M 
357 212 
```

Q3. How many variables/features in the data are suffixed with _mean?

```r
colnames(wisc.df)
```

```
 [1] "diagnosis"               "radius_mean"            
 [3] "texture_mean"            "perimeter_mean"         
 [5] "area_mean"               "smoothness_mean"        
 [7] "compactness_mean"        "concavity_mean"         
 [9] "concave.points_mean"     "symmetry_mean"          
[11] "fractal_dimension_mean"  "radius_se"              
[13] "texture_se"              "perimeter_se"           
[15] "area_se"                 "smoothness_se"          
[17] "compactness_se"          "concavity_se"           
[19] "concave.points_se"       "symmetry_se"            
[21] "fractal_dimension_se"    "radius_worst"           
[23] "texture_worst"           "perimeter_worst"        
[25] "area_worst"              "smoothness_worst"       
[27] "compactness_worst"       "concavity_worst"        
[29] "concave.points_worst"    "symmetry_worst"         
[31] "fractal_dimension_worst"
```

```r
grep("_mean", colnames(wisc.df))
```

```
 [1]  2  3  4  5  6  7  8  9 10 11
```

```r
length(grep("_mean", colnames(wisc.df)))
```

```
[1] 10
```

We need to remove the `diagnosis` column before we do any further analysis of this data set - we don't want to pass this to PCA etc. We will save it as a separate vector that we can use later to compare our findings to those of experts.

```r
wisc.data <- wisc.df[, -1]
diagnosis <- wisc.df$diagnosis
```

Principal Component Analysis (PCA)

The main function in base R is called `prcomp()`. We will use the optional arguments `scale=TRUE` here as the data columns/features/dimensions are on very different scales in the origianl data set.

```r
wisc.pr <- prcomp(wisc.data, scale=TRUE)
```

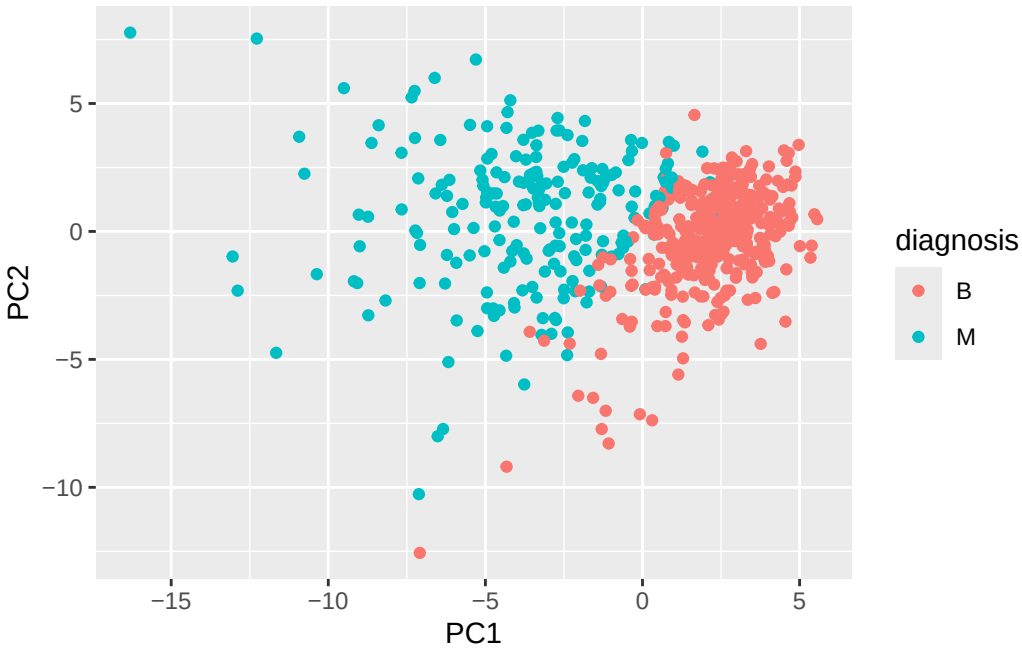
```r
attributes(wisc.pr)
```

```
$names
[1] "sdev"     "rotation" "center"   "scale"    "x"

$class
[1] "prcomp"
```

```r
library(ggplot2)

ggplot(wisc.pr$x) +
  aes(PC1, PC2, col=diagnosis) +
  geom_point()
```




```r
summary(wisc.pr)
```

```
Importance of components:
                          PC1    PC2     PC3     PC4     PC5     PC6     PC7
Standard deviation     3.6444 2.3857 1.67867 1.40735 1.28403 1.09880 0.82172
Proportion of Variance 0.4427 0.1897 0.09393 0.06602 0.05496 0.04025 0.02251
Cumulative Proportion  0.4427 0.6324 0.72636 0.79239 0.84734 0.88759 0.91010
                           PC8    PC9    PC10   PC11    PC12    PC13    PC14
Standard deviation     0.69037 0.6457 0.59219 0.5421 0.51104 0.49128 0.39624
Proportion of Variance 0.01589 0.0139 0.01169 0.0098 0.00871 0.00805 0.00523
Cumulative Proportion  0.92598 0.9399 0.95157 0.9614 0.97007 0.97812 0.98335
                          PC15    PC16    PC17    PC18    PC19    PC20   PC21
Standard deviation     0.30681 0.28260 0.24372 0.22939 0.22244 0.17652 0.1731
Proportion of Variance 0.00314 0.00266 0.00198 0.00175 0.00165 0.00104 0.0010
Cumulative Proportion  0.98649 0.98915 0.99113 0.99288 0.99453 0.99557 0.9966
                          PC22    PC23   PC24    PC25    PC26    PC27    PC28
Standard deviation     0.16565 0.15602 0.1344 0.12442 0.09043 0.08307 0.03987
Proportion of Variance 0.00091 0.00081 0.0006 0.00052 0.00027 0.00023 0.00005
Cumulative Proportion  0.99749 0.99830 0.9989 0.99942 0.99969 0.99992 0.99997
                          PC29    PC30
Standard deviation     0.02736 0.01153
Proportion of Variance 0.00002 0.00000
Cumulative Proportion  1.00000 1.00000
```

Q4. From your results, what proportion of the original variance is captured by the first principal component (PC1)?

```
summary(wisc.pr)$importance[2, 1]
```

```
[1] 0.44272
```

Q5. How many principal components (PCs) are required to describe at least 70% of the original variance in the data?

```
table(summary(wisc.pr)$importance[3,] >= 0.70)
```

```
FALSE  TRUE
    2    28
```

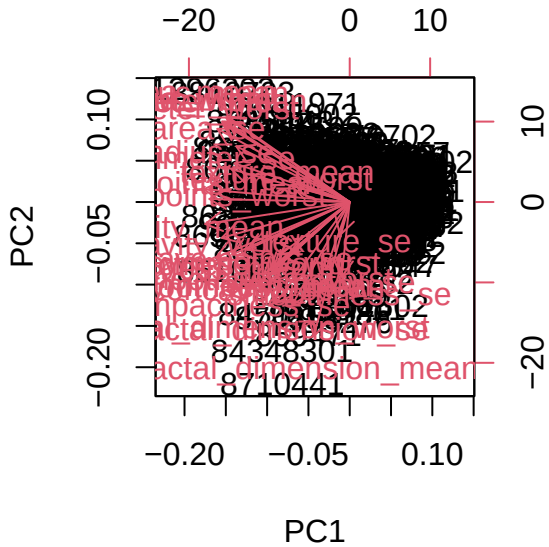
Q6. How many principal components (PCs) are required to describe at least 90% of the original variance in the data?

```
table(summary(wisc.pr)$importance[3,] >= 0.90)
```

```
FALSE  TRUE
    6    24
```

Q7. What stands out to you about this plot? Is it easy or difficult to understand? Why?

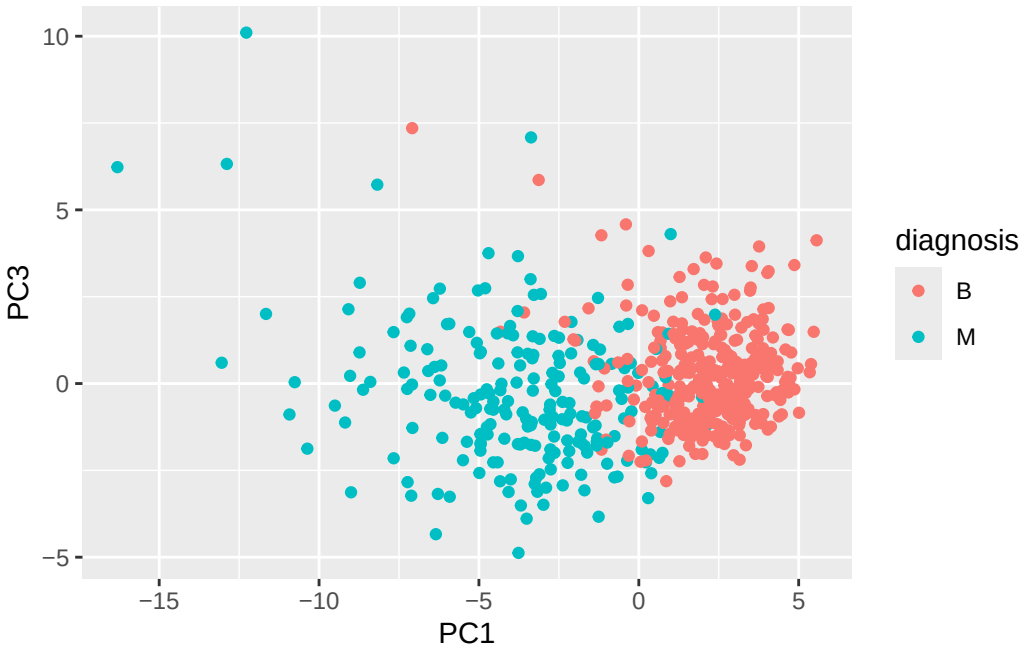
```
biplot(wisc.pr)
```



- The plot is messy and hard to understand, in that the data is very clustered together to where it is difficult to read each individual label and difficult to understand what the plot is actually trying to say. The plot also has numbers on all sides of the graph so it isn't as clear to see which values are actually being compared to one another.

  Q8. Generate a similar plot for principal components 1 and 3. What do you notice about these plots?

```
ggplot(wisc.pr$x) +
  aes(PC1, PC3, col=diagnosis) +
  geom_point()
```



- The plot has an easy distinction between the variables and each individual patient can easily be seen and determined whether they are benign or malignant. It is also clear easier to see how the two different diagnosis compare to one another in value in that "M" occupies the left half of the graph and "B" the right half.

  Q9. For the first principal component, what is the component of the loading vector (i.e. wisc.pr$rotation[,1]) for the feature concave.points_mean? This tells us how much this original feature contributes to the first PC. Are there any features with larger contributions than this one?

```
wisc.pr$rotation[,1]
```

```
            radius_mean           texture_mean         perimeter_mean
            -0.21890244            -0.10372458            -0.22753729
              area_mean        smoothness_mean       compactness_mean
            -0.22099499            -0.14258969            -0.23928535
         concavity_mean    concave.points_mean          symmetry_mean
            -0.25840048            -0.26085376            -0.13816696
 fractal_dimension_mean              radius_se             texture_se
            -0.06436335            -0.20597878            -0.01742803
           perimeter_se                area_se          smoothness_se
            -0.21132592            -0.20286964            -0.01453145
         compactness_se           concavity_se      concave.points_se
```

```
        -0.17039345            -0.15358979            -0.18341740
        symmetry_se    fractal_dimension_se           radius_worst
        -0.04249842            -0.10256832            -0.22799663
      texture_worst         perimeter_worst             area_worst
        -0.10446933            -0.23663968            -0.22487053
    smoothness_worst       compactness_worst        concavity_worst
        -0.12795256            -0.21009588            -0.22876753
concave.points_worst          symmetry_worst fractal_dimension_worst
        -0.25088597            -0.12290456            -0.13178394
```

- The component of the loading vector for `concave.points_mean` is -0.26, in which the negative sign means it decreases the value of the PC. Given that there is no other value that is bigger than `concave.points_mean`, this loading vector is the largest contributor.

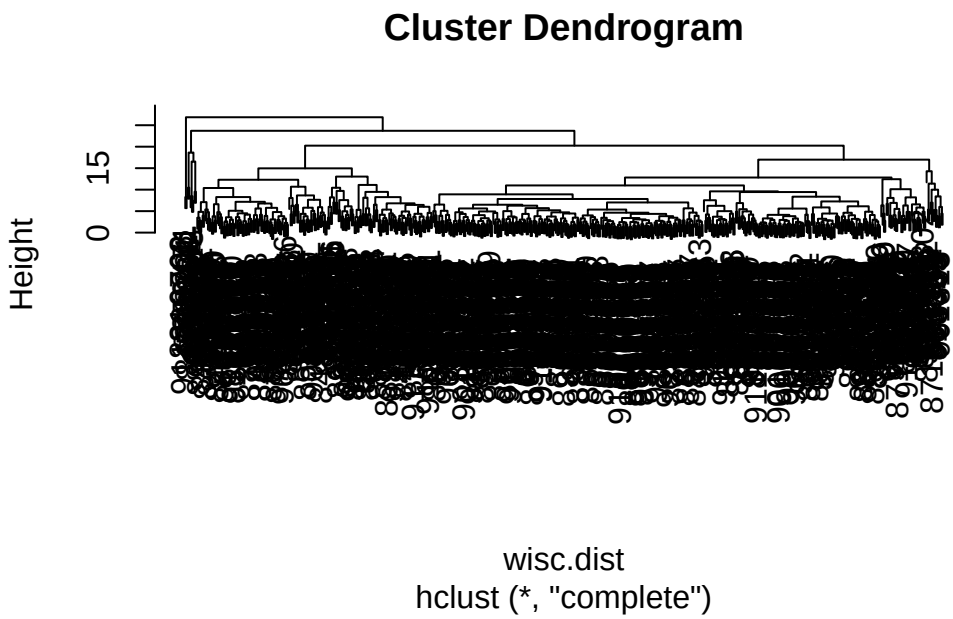
Hierarchical Clustering

The goal of this section is to do hierarchical clustering of the **original data** to see if there is any obvious grouping into malignant and benign clusters.

The results are not good

First, we will scale our `wisc.data` then calculated a distance matrix, then pass to `hclust()`:

```r
wisc.dist <- dist(scale(wisc.data))
wisc.hclust <- hclust(wisc.dist)
plot(wisc.hclust)
```

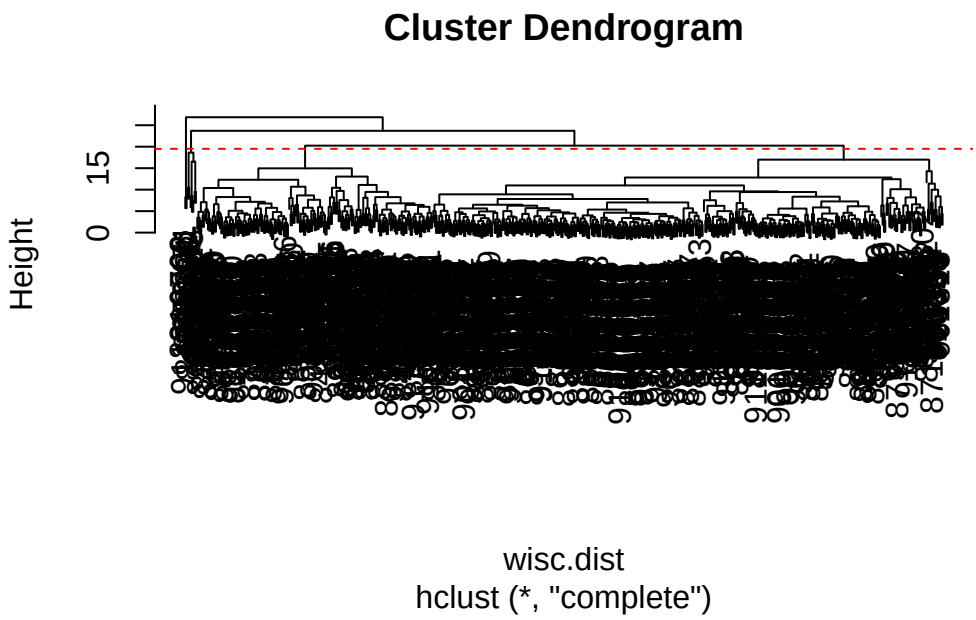



```
wisc.hclust.clusters <- cutree(wisc.hclust, k=2)
table(wisc.hclust.clusters)
```

```
wisc.hclust.clusters
  1   2
567   2
```

Q10. Using the plot() and abline() functions, what is the height at which the clustering model has 4 clusters?

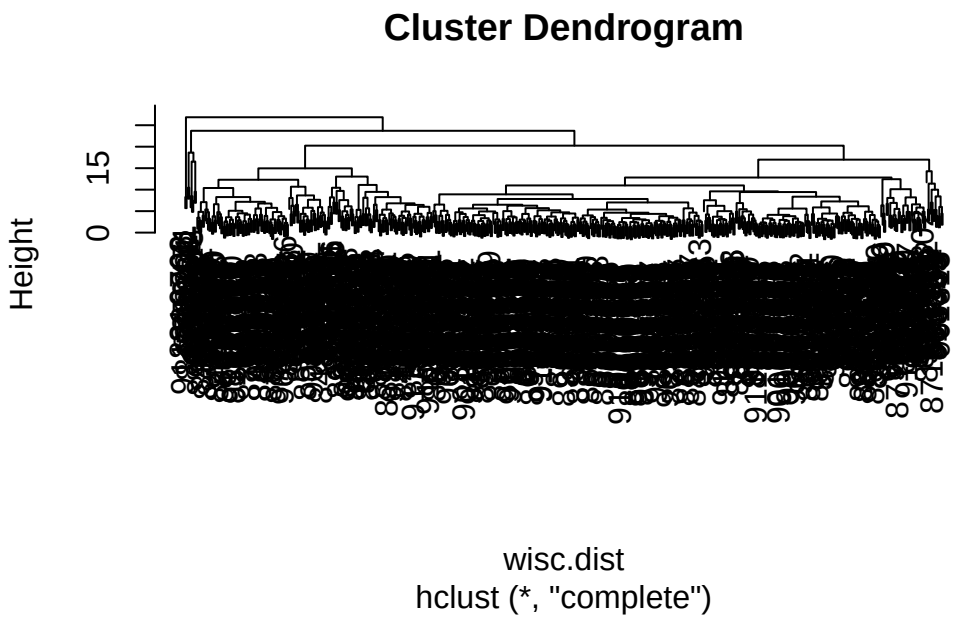
```
plot(wisc.hclust)
abline(h=19.5, col="red", lty=2)
```



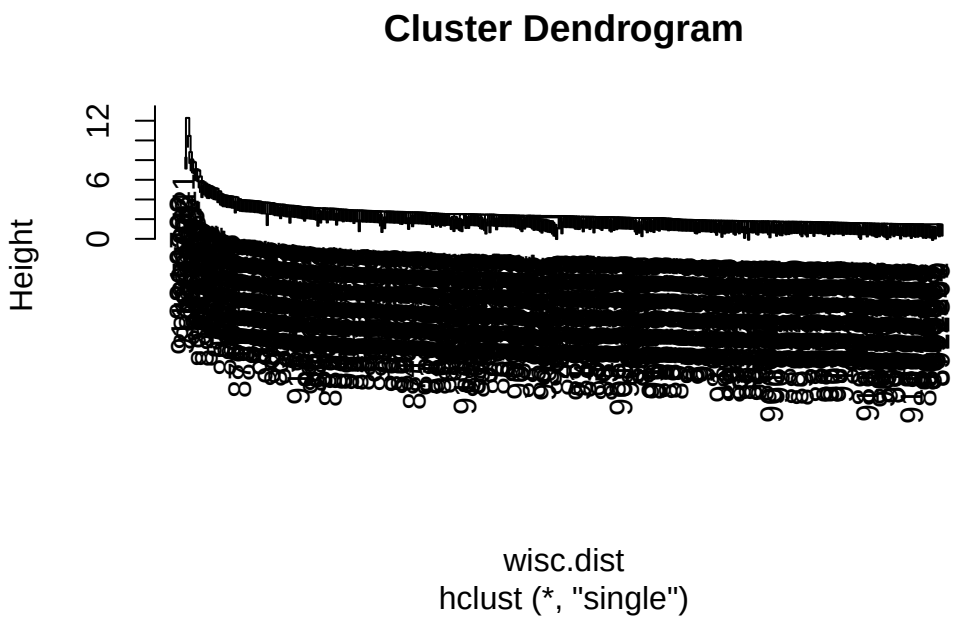
Q12. Which method gives your favorite results for the same `data.dist()` dataset?

- The method "ward.D2" has favored results as it is less messy and crowded to look at, in that the clusters can be easily seen from orrignating from the first cluster. The second would be the method "complete" for the same reasons but there are more clusters that are produced, which can be good or not as needed depending on what is looked for

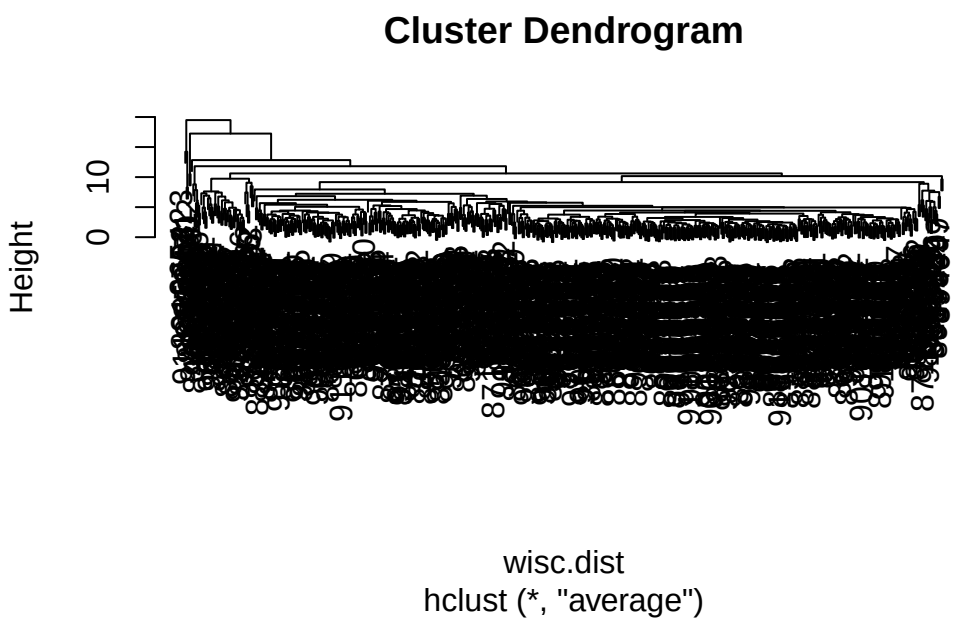
```
hc.complete <- hclust(wisc.dist, method="complete")
plot(hc.complete)
```



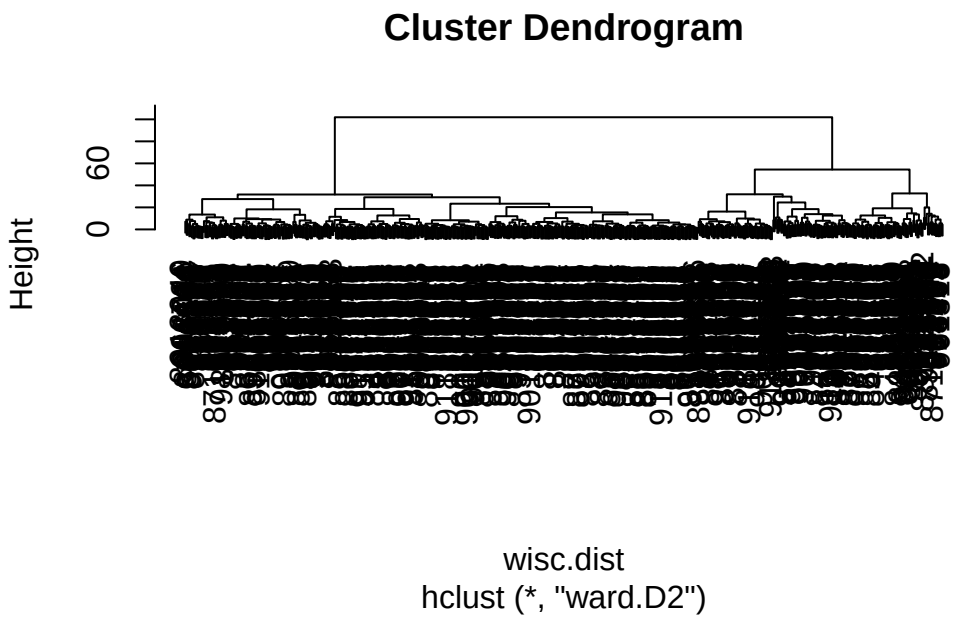

```
hc.single <- hclust(wisc.dist, method="single")
plot(hc.single)
```



```r
hc.average <- hclust(wisc.dist, method = "average")
plot(hc.average)
```



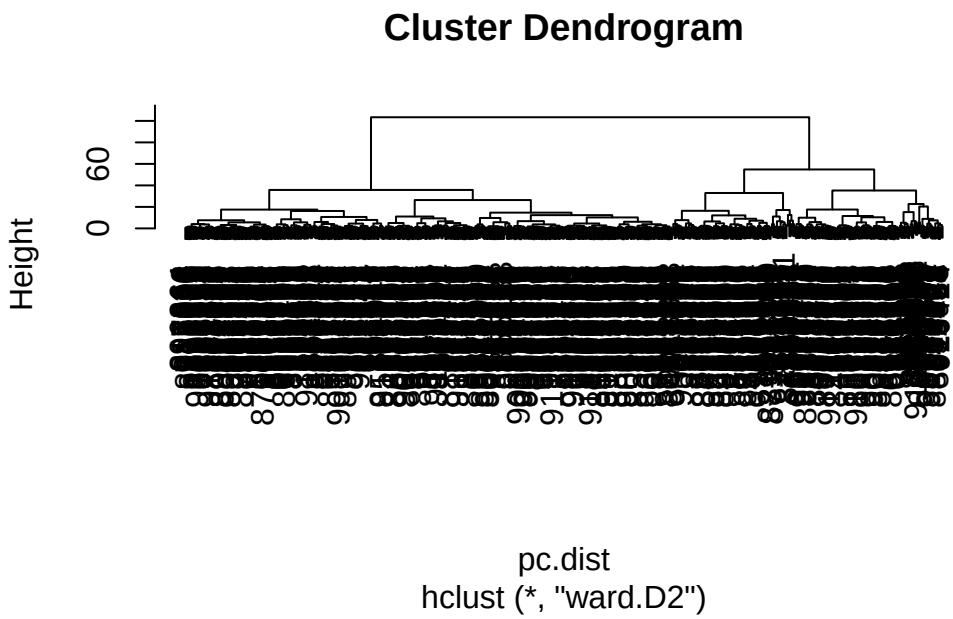

```r
hc.ward <- hclust(wisc.dist, method="ward.D2")
plot(hc.ward)
```



Combining Methods

The idea here is that I can take my new variables (the scores on the PCs `wisc.pr$x`) that are better descriptors of the data set than the original features (i.e. the 30 columns in `wisc.data`) and use these as a basis for clustering.

```
pc.dist <- dist(wisc.pr$x[,1:3])
wisc.pr.hclust <- hclust(pc.dist, method="ward.D2")
plot(wisc.pr.hclust)
```




```
grps <- cutree(wisc.pr.hclust, k=2)
table(grps)
```

```
grps
  1   2
203 366
```

```
table(diagnosis)
```

```
diagnosis
  B   M
357 212
```

I can now run `table()` with both my clustering `grps` and the expert `diagnosis`

> Q13. How well does the newly created `hclust()` model with the two clusters separate out the two "M" and "B" diagnoses?

```
table(grps, diagnosis)
```

```
    diagnosis
grps   B   M
   1  24 179
   2 333  33
```

> Q14. How well do the hierarchical clustering models created do in terms of separating the diagnoses

Our cluster "1" has 179 "M" diagnosis

- True Positive (TP): 179 False Positive (FP): 24

Our cluster "2" has 333 "B" diagnosis

- True Negative (TN): 333
- False Negative (FN): 33

```r
ftable(grps, wisc.hclust.clusters, diagnosis)
```

```
                          diagnosis   B   M
grps wisc.hclust.clusters
1    1                               24 177
     2                                0   2
2    1                              333  33
     2                                0   0
```

Sensitivity + Specificity

Sensitivity: TP/(TP+FN)

```r
179/(179+33)
```

```
[1] 0.8443396
```

Perfect: Sensitivity of 1

Specificity: TN/(TN+FP)

```r
333/(333+24)
```

```
[1] 0.9327731
```

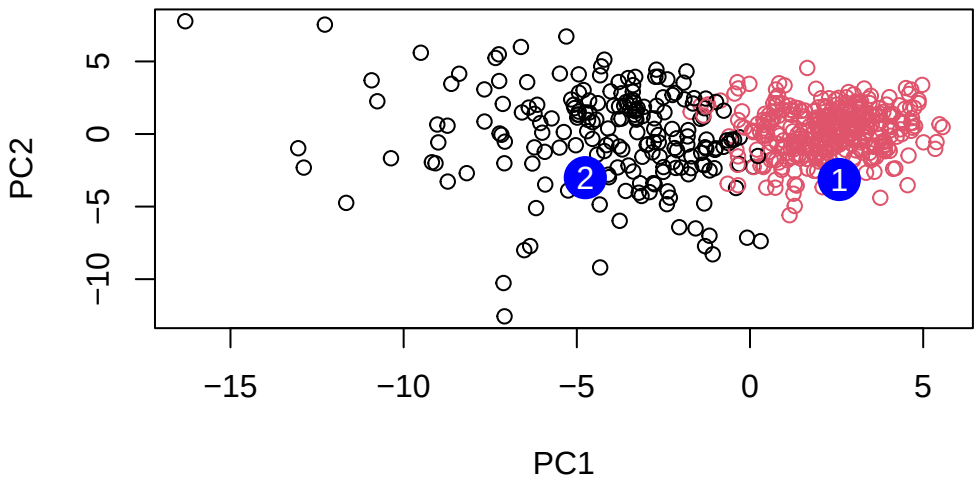
Prediction

We can use our PCA model for prediction of new un-seen cases:

```
url <- "https://tinyurl.com/new-samples-CSV"
new <- read.csv(url)
npc <- predict(wisc.pr, newdata=new)
npc
```

```
           PC1       PC2        PC3        PC4       PC5        PC6        PC7
[1,]  2.576616 -3.135913  1.3990492 -0.7631950  2.781648 -0.8150185 -0.3959098
[2,] -4.754928 -3.009033 -0.1660946 -0.6052952 -1.140698 -1.2189945  0.8193031
            PC8       PC9       PC10      PC11      PC12      PC13     PC14
[1,] -0.2307350 0.1029569 -0.9272861 0.3411457  0.375921 0.1610764 1.187882
[2,] -0.3307423 0.5281896 -0.4855301 0.7173233 -1.185917 0.5893856 0.303029
          PC15       PC16        PC17        PC18        PC19       PC20
[1,] 0.3216974 -0.1743616 -0.07875393 -0.11207028 -0.08802955 -0.2495216
[2,] 0.1299153  0.1448061 -0.40509706  0.06565549  0.25591230 -0.4289500
           PC21       PC22       PC23       PC24        PC25         PC26
[1,]  0.1228233 0.09358453 0.08347651  0.1223396  0.02124121  0.078884581
[2,] -0.1224776 0.01732146 0.06316631 -0.2338618 -0.20755948 -0.009833238
             PC27        PC28         PC29         PC30
[1,]  0.220199544 -0.02946023 -0.015620933  0.005269029
[2,] -0.001134152  0.09638361  0.002795349 -0.019015820
```

```
plot(wisc.pr$x[,1:2], col=grps)
points(npc[,1], npc[,2], col="blue", pch=16, cex=3)
text(npc[,1], npc[,2], c(1,2), col="white")
```



Q16. Which of these new patients should be prioritized for follow up based on the results

- Patient two should be prioritized as its PC1 value of -5 means that it is a greater influence to the value of the PC. Based on the previous graphs, patient two would have been diagnosed with a malignant tumor whereas patient one would have been diagnosed with a benign tumor.